COMPARISION OF MACHINE LEARNING ALGORITHMS IN PREDICTING HEART DISEASES

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ABSTRACT:

This Heart diseases study predicting whether or not someone has heart disease based on their medical attributes using machine learning. With a dataset of 303 samples and 14 variables, including trestbps and chol, Linear Regression, Decision Tree, and Random Forest models are employed. Rigorous pre-processing ensures data cleanliness. Visualizations, like box plots and heat maps, reveal insights into distribution and correlations. Through a 7-fold cross-validation, Decision Tree emerges as the top performer, emphasizing its efficacy for small to medium-sized datasets in energy prediction. This Researching machine learning algorithms to predict heart disease risk factors based on demographic, lifestyle, and medical data, aiming to enhance early detection and prevention strategies. Future research involves exploring additional algorithms and parameter tuning for enhanced accuracy.

Keywords and Phrases: Machine Learning, Linear Regressor, Decision Tree, Random Forest, Python, Heart diseases.

1 INTRODUCTION

Shruthi K et al [1] focuses on the matter of heart diseases as a significant global public health concern highlighting the difficulties, in predicting and preventing them due to various factors. The study adopts a machine learning approach utilizing algorithms and a wide range of healthcare data sources to improve Heart Diseases risk assessment and prevention achieving a noteworthy accuracy rate of 96.9%. Techniques employed include data preparation, feature selection, model creation using SVM, KNN and Random Forest algorithms and assessment through confusion matrices and k fold cross validation. The research involves data gathering, cleansing and analysis using Python with tools like pandas and NumPy. The study proposes a system for predicting heart health that stresses the importance of detection and intervention. It highlights the value of interpretability and ethical considerations while emphasizing the benefits of cooperation in improving risk assessment and intervention strategies. Positive Outcomes obtained accuracy showcases the impact of machine learning on enhancing heart management; however further exploration into the complexities of CVDs along, with incorporating variables is recognized as necessary. Challenges may arise from issues related to data quality as the risk of model overfitting. This research is valuable for healthcare experts, policymakers and the general public as it provides information, on enhancing disease prediction and prevention. It highlights how machine learning can support healthcare initiatives and lessen the impact of diseases. The research conducted by Jyothirmai D et al [2] discusses improving the accuracy of predicting diseases (CVDs) using machine learning techniques such, as Logistic Regression, Random Forests and k nearest neighbors (KNN). The main goal is to create a model that can predict heart diseases accurately with a set of input factors. The methodology used is machine learning algorithms on a dataset of 303 patients with 13 attributes. Hyper parameter optimization methods are used to improve efficiency. The key discovery is that the Support Vector Machine technique demonstrates performance in terms of accuracy and effectiveness compared to approaches. The research includes an examination of machine learning techniques, dataset preparation methods, feature selection strategies and performance assessment metrics. Tools and tactics encompass regression, random forests, k nearest neighbors, hyper parameter optimization well as feature selection approaches like LASSO and relief optimization. In conclusion the Random Forest algorithm surpasses KNN and logistic regression by achieving a precision rate of 94%. Effective feature selection significantly enhances performance. The study underscores the potential of machine learning in predicting CVDs and refining medical diagnostic processes. Encouraging outcomes highlight the identification of traits, through the high precision rate achieved by Random Forest. However, like [1] this study results hold significance indicate that advanced machine learning approaches could notably enhance heart disease prediction accuracy. This has implications for healthcare hinting at potential progress, in early disease detection and personalized medicine that could ultimately result in improved patient outcomes. For instances, A Solanki et al [3] focuses on the challenge of predicting heart disease using machine learning techniques to enhance outcomes through timely intervention and therapy. Various machine learning models, including Logistic Regression and Decision Tree Classifier were utilized in the study, which was based on an dataset, on heart disease. The Decision Tree Classifier emerged as the model for forecasting cardiac

disease showcasing its ability to capture complex data relationships. Data preprocessing steps involved managing missing data encoding variables and standardizing features. Feature selection methods were used to identify predictors. Evaluation criteria such as accuracy, precision and recall were employed to evaluate model performance. The Decision Tree Classifier proved to have the accuracy among all models tested showing promise for prediction of heart disease. Nonetheless limitations related to representation and model interpretability were observed, emphasizing the need for research and enhancement of models. While the study underscores the potential of machine learning in predicting heart disease it also acknowledges limitations such as dataset biases and challenges in interpreting models. This calls for caution, in model interpretation. Suggests incorporating clinical factors to enhance accuracy. Strengths and weaknesses of the study the research highlights the effectiveness of the approach used and the development of a precise predictive model. However, limitations, in the dataset and challenges in understanding models may impact their use in clinical settings. The research sets a groundwork for exploration of machine learning in health leading to advancements in risk assessment, early detection and personalized treatment strategies. These discoveries have reaching effects on enhancing analytics, in healthcare and enhancing patient outcomes. Similarly, Sunit Mistry et al [4] discusses the challenge of identifying disease in the presence of common conditions, like diabetes and hypertension. It highlights the importance of handling issues due to their complexity and potential negative effects on patients. In terms of methodologies and significant findings the research utilizes a range of data mining techniques. One key discovery is the development of an approach (HRFLM) that combines Random Forest with Linear Method showing results in predicting cardiac disease with an accuracy rate of 89%. For research implementation the study utilizes the Cleveland UCI repository dataset for classifying disease and employs R Studio Rattle for analysis purposes. The data processing involves feature extraction and selection followed by classification modeling using machine learning algorithms. The study concludes that early detection of disease through machine learning methods can lead to a reduction in mortality rates. Broader Impact; The study highlights how machine learning has potential to transform cardiac disease diagnosis and treatment emphasizing the value of collaboration between medical professionals and data scientists for significant advancements, in healthcare. Anand kumar Shukla [5] delves into the use of machine learning to predict diseases supporting healthcare professionals in developing care strategies. By employing machine learning algorithms such, as K Nearest Neighbor on a database related to Cleveland Heart Disease the study highlights the efficacy of algorithms in predicting risks. The research involves data preprocessing the application of classification algorithms and performance assessment using precision scores. It concludes that key attributes play a role in accurate disease risk prediction with promising outcomes observed for algorithms like K Nearest Neighbor and Naïve Bayes. While achieving precision scores is notable there is a call for sophisticated models and diverse datasets to enhance accuracy further. The strengths lie in the variety of algorithms used and attribute focused predictions while limitations are attributed to constraints and model intricacies. His research holds significance for healthcare by improving risk assessment and early disease detection for patient outcomes and cost savings in healthcare. It emphasizes the importance of data mining in healthcare settings advocating its integration into practices, for efficient disease management and prevention.

2 METHODOLOGY USED

The methodology of this project is explained in the below sub-sections.

2.1 Dataset collection

Many datasets were available for performing this project but the dataset for this project is collected from Kaggle website with the name of the dataset “Energy Efficiency Data Set” which seems applicable to the project outcomes. The size of the dataset is 303 x 14 i.e., 303 rows and 14 columns or variables in which 3 variables are dependent variables and 8 variables are independent variables. The dependent variables are sex and age. This dataset is downloaded in csv format and Jupyter notebook is used to perform the python code on the dataset.

2.2 Data preprocessing

Before working on any dataset, the primary step is to clean the dataset using data cleaning techniques such as handling missing values, handling null values, handling duplicate values, handling outliers, handling feature engineering, and data normalization or standardization. In the dataset taken, the dataset is pretty cleaned, there are no null values, no missing values, no duplicate values, and no outliers as well. Feature engineering is done by removing dependent variables separately to perform machine learning algorithms. Hence, the dataset has quality data to work up on.

2.3 Data visualization

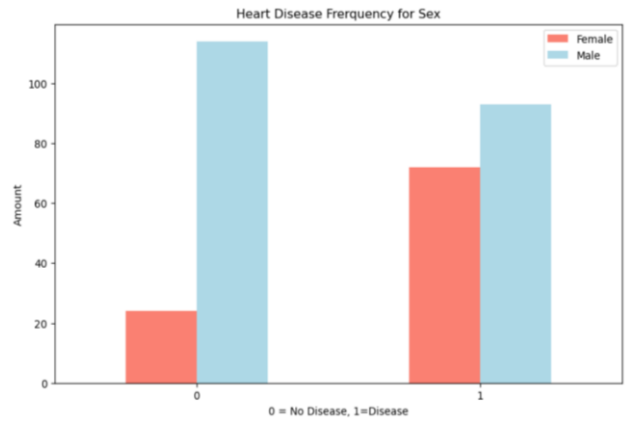
Data visualization is an important step in the methodology. It is a pictorial representation of the data such that relevant findings can be observed easily. This data visualization helps in enhancing understanding and supporting decision making. With the help of data visualization, complex data can be transformed to clear representations (Sunit Mistry et al [4]). There are many libraries such as matplotlib, seaborn, plotly etc., in this project, seaborn and matplotlib is used to generate box plot, scatter plot, histplot, countplot, bar chart and heatmap etc., are used for interactive visualization of the data present in the dataset. For instance, using box plot for checking outliers in each feature or variable helped in identifying that no outliers were present in the dataset. Similarly, histplot is used to observe the variation of sex, Cholesterol and chest pain values in the dataset. Likewise, remaining plots are generated for understanding different entities of the data.

2.4 Model selection and Performance evaluation

In this project, three regression models have been chosen to perform prediction on the dataset. The three algorithms considered are Decision Tree Regressor, Random Forest Regressor and LinearRegressor. These three models were chosen because these three models better work on small to medium size datasets compared to algorithms such as Gradient Boost, AdaBoost, and XG Boost. These regression models were put under 7 folds cross validation. For each of the fold, metrics such as R2 score and mean squared error (MSE) are chosen for calculating the dependent variables. These metrics were only chosen because these are easy to implement and interpret the results and is also sufficient for the project scope for this dataset. The dataset is trained and tested with ratio of 3:5 for training data to testing data. Feature importance is identified for all three of the models. Finally, ranking of the algorithm is done based on the R2 score evaluation. The reason behind choosing R2 score is because it gives a comprehensive idea of prediction of all three of the models.

3 RESULTS AND DISCUSSIONS

After completion model testing and performance evaluation, there are some notable findings from the dataset observed. Firstly, the dataset is clean with no missing values, no outliers, no duplicate values, no missing values, and feature engineering is done by separating dependent variables sex and age. The variation in the dependent variable values is observed using histplot.



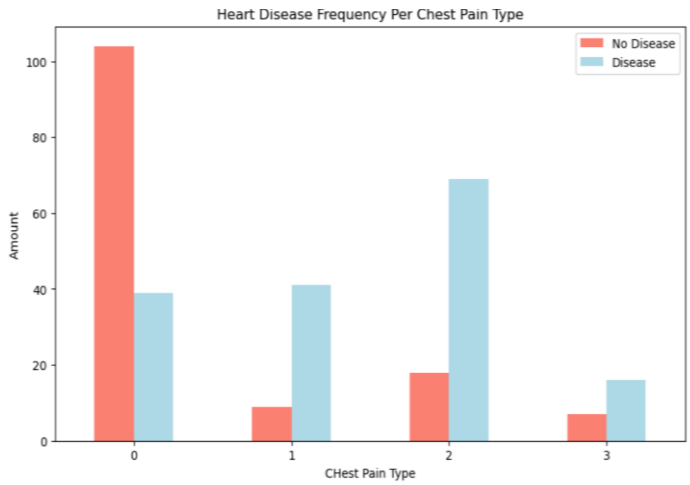
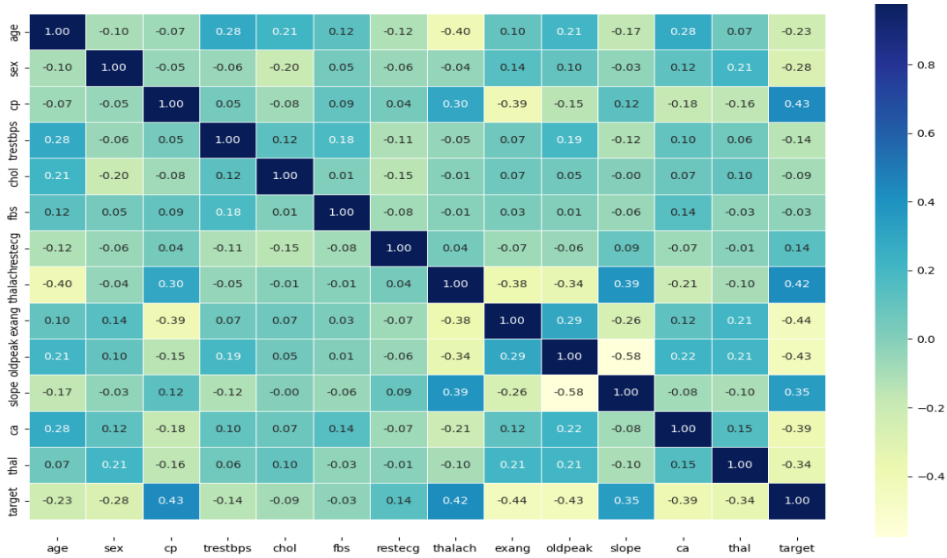


Figure 2: Histplot of the dependent variables

Figure 2 explains that there are variations in the values implying that there is no one-side tendency of values which means the dataset can be used for prediction models. The class distribution is checked using countplot for the target variables. Correlation between the variables of the dataset in understood by the help of figure 3 which can help in understanding the relationships, strengths, and direction of associations among the variables.



From figure 3, it is observed that Orientation, Glazing_area, Glazing_area_distribution are having no corelation with many of the variables and sex, Chest pain are highly correlated with Overall_height of the building. Later, a 7-fold cross validation technique is applied to three algorithms and it is observed that random forest took less time to complete the iterations whereas linear regressor was last to complete. Feature importance to prediction of models is done for each machine learning model.

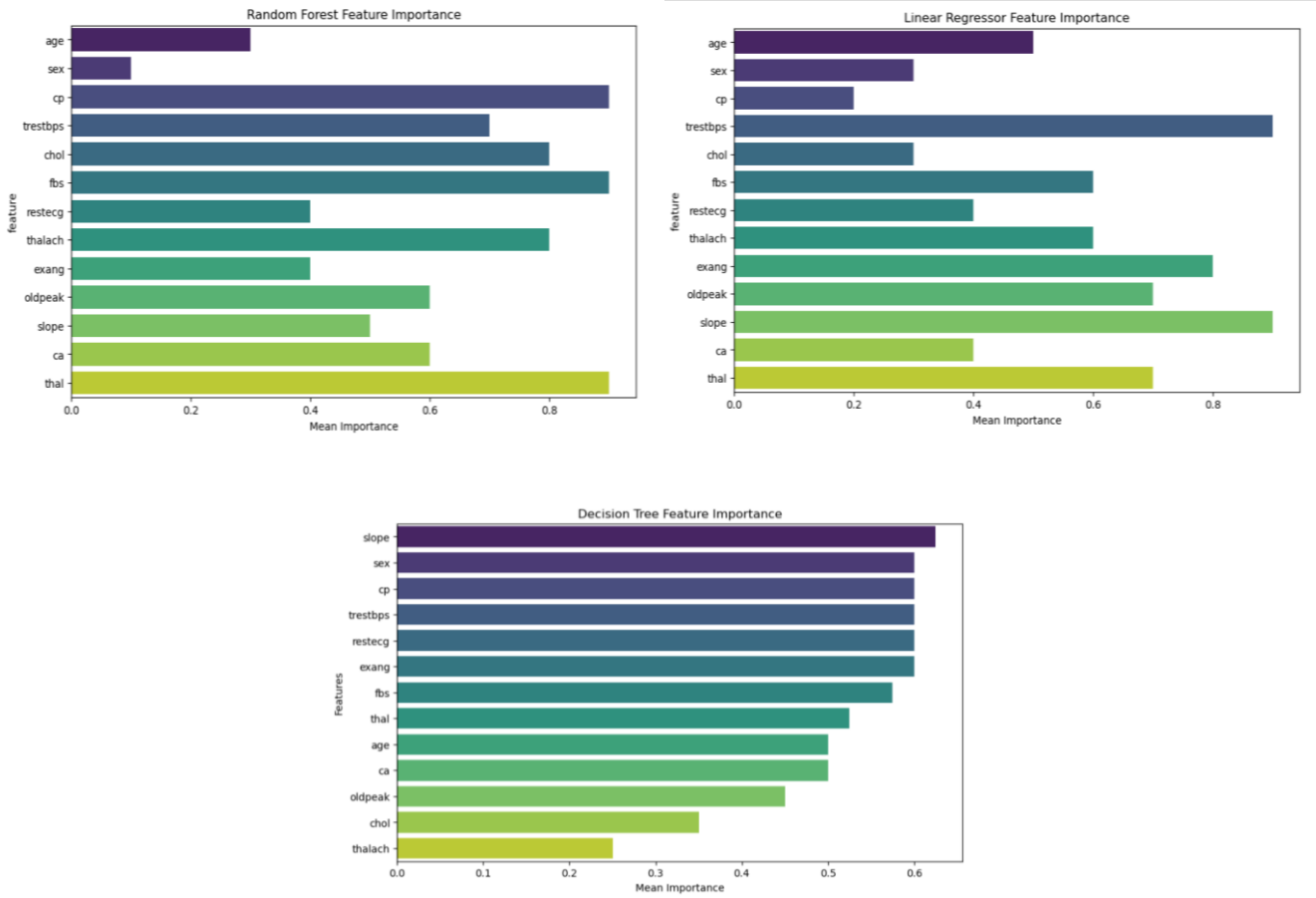
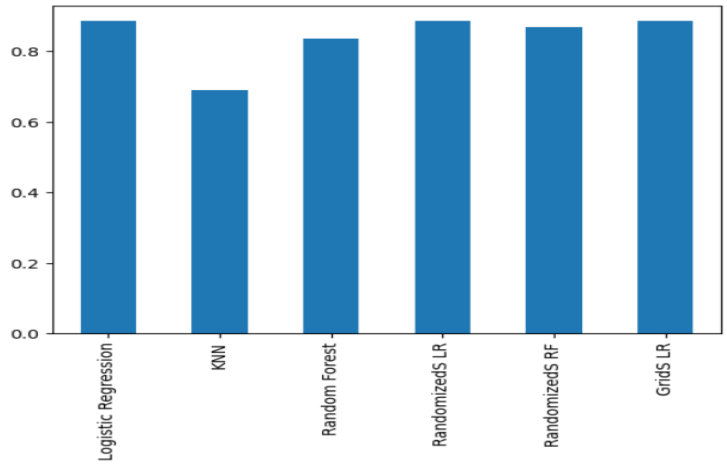


Figure 4: Feature importance according to each algorithm

Figure 4 indicates that for all the three regressors shows Relative Compactness obtains to be most important feature whereas Linear regressor shows negative importance to Orientation variable.




```
precision    recall  f1-score   support
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           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
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Figure 6: Ranking of algorithms

The figure 5 and figure 6 shows that Random Forest regressor best fits to the dataset for prediction of target variables based on R2 score ranking. The decision tree regressor stands next to it with a little difference in accuracy. While the linear regressor attains last rank and appears to be unsuitable fit for the dataset comparatively. The random forest worked better because the random forest comparatively with other regressor algorithms has better performance due to ensemble learning and variance reduction. While comparing with other literature, (Jyothirmai D et al., 2021) describes AdaBoost works better or performs better on large datasets, (Anand kumar Shukla., 2016) describes Gradient Boost are better workable on large datasets with binary or multiclass classification. The use of Supervised Regressor can be taken into consideration for this kind of dataset as it best works on labelled data (Sunit Mistry et al, 2016).

4 CONCLUSION AND FUTURE WORKS

In conclusion, upon performing the three machine learning algorithms on the dataset which is considered as small to medium sized dataset, random forest regressor obtained to be best performed algorithm with a R2 score of 99.6% and following random forest regressor, decision tree regressor obtained 99.5% and linear regressor scored last with 91.1% on age variable. Similarly, random forest regressor obtained to be best performed algorithm with a R2 score of 96% and following random forest regressor, decision tree regressor obtained 94.1% and linear regressor scored last with 88.1% on sex variable. It is because the random forest regressor best works because of ensemble learning and variance reduction optimization present in the algorithm. The future works of this project can be implementing various other algorithms and also parameter tuning can be done to observe different results. Additionally, performing the same operations on a large dataset can be done to check the model generalizability in the real-world scenarios.

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